

1. Introduction:

1.1 Background

The aviation industry is a critical pillar of global transportation, facilitating the rapid movement of people and resources. Within the United States, domestic air travel constitutes a massive and highly complex logistical network. Ensuring the smooth operation of this network requires precise scheduling, rigorous safety protocols, and effective time management.

1.2 Significance of the Problem

Despite advanced scheduling systems, flight delays remain a persistent challenge, leading to significant economic inefficiencies, passenger dissatisfaction, and operational disruptions. Managing and understanding these delays relies heavily on massive volumes of operational data. However, the raw data recorded by aviation authorities often contains inconsistencies, missing values, and errors. Addressing these data quality issues is an essential prerequisite; without a clean, reliable dataset, any subsequent analysis or predictive modelling will be fundamentally flawed, leading to inaccurate conclusions and ineffective solutions.

1.3 Research Question and Objectives This study bridges the gap between raw aviation data and actionable insights through applied Data Science. The primary objective is to answer the following research question: *"What are the principal causes and trends of flight delays for U.S. domestic air carriers, and how might improvements to data quality reduce the uncertainties associated with the analysis of on-time performance using BTS reporting data?"*. Specifically, this research aims to build a robust data pipeline to resolve data quality issues, analyse delay patterns, and ultimately develop a machine learning binary classification model to predict the likelihood of flight delays.

1.4 Structure of the Paper

The remainder of this paper is structured as follows: Section 2 details the dataset and the data integration process. Section 3 outlines the methodology, encompassing the data cleaning pipeline and Exploratory Data Analysis (EDA). Section 4 presents the development and evaluation of the machine learning models (Logistic Regression, Random Forest, and XGBoost). Finally, Section 5 discusses the expected outcomes, limitations, and potential avenues for future research.

2. Literature Review:

2.1 Overview of Flight Delay Predictions

Flight delay prediction has been extensively studied within transportation analytics due to the complex logistical network of domestic air travel. Historically, researchers relied on macroscopic statistical models to characterize delay distributions. However, as demonstrated by Rebollo and Balakrishnan [1], the shift towards Machine Learning (ML) has enabled the capture of complex temporal and spatial patterns within aviation networks. Their foundational work highlighted that predictive modelling can significantly reduce uncertainties in air traffic management by treating delays as classification and regression problems.

2.2 Key Methodologies and Findings

Recent literature has increasingly focused on leveraging ensemble learning algorithms to process massive datasets, such as those provided by the Bureau of Transportation Statistics (BTS). For instance, Choi et al. [2] evaluated multiple algorithms, including Random Forest, to predict weather-induced airline delays, emphasizing the model's robustness and feature importance interpretation. Furthermore, a comprehensive study by Gui et al. [3] utilized advanced tree-based models, notably eXtreme Gradient Boosting (XGBoost) and LightGBM, on aviation big data. Their findings proved that XGBoost achieves superior tabular performance in binary classification tasks compared to baseline models like Logistic Regression, directly supporting the algorithmic choices outlined in this research proposal.

2.3 Research Gap: The Role of Data Quality

While the mentioned studies successfully apply sophisticated predictive algorithms [1]–[3], a critical gap remains regarding the rigorous management of data quality prior to modelling. Many existing papers either assume a pristine dataset or document only superficial preprocessing steps. They often overlook the profound impact that structural inconsistencies, missing values, and anomalies have on the reliability of on-time performance analysis. This research explicitly addresses this gap by developing a comprehensive data quality pipeline. By systematically resolving these issues before deploying baseline and advanced models, this study aims to demonstrate that stringent data integration and cleaning are indispensable for achieving highly accurate and actionable flight delay predictions.

References:

[1] J. J. Rebollo and H. Balakrishnan, "Characterization and prediction of air traffic delays," *Transportation Research Part C: Emerging Technologies*, vol. 44, pp. 231-241, 2014.

Available: <https://doi.org/10.1016/j.trc.2014.04.007>

[2] S. Choi, Y. J. Kim, S. Briceno, and D. Mavris, "Prediction of weather-induced airline delays based on machine learning algorithms," in *2016 IEEE/AIAA 35th Digital Avionics Systems Conference (DASC)*, Sacramento, CA, USA, 2016, pp. 1-6. Available:

<https://doi.org/10.1109/DASC.2016.7778090>

[3] G. Gui, F. Liu, J. Sun, J. Yang, Z. Zuo, and S. Shen, "Flight Delay Prediction Based on Aviation Big Data and Machine Learning," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 1, pp. 140-150, Jan. 2020. Available: <https://doi.org/10.1109/TVT.2019.2954066>